

ON FINITE AND INFINITE CONFIDENCE SEQUENCES

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SUMMARY

In this paper we study nonparametric finite confidence sequences for the median of a distribution and for a regression parameter. The results of some Monte Carlo experiments are reported. Examples of the application of infinite confidence sequences to some problems of statistical inference are also given.

1. Introduction

Let P_θ be a probability measure under which the sequence $\{X_i, i \geq 1\}$ of r.v.s. are independently distributed with c.d.f.'s $\{F_{\theta,i}, i \geq 1\}$. A sequence of intervals $\{I_n \equiv I_n(X_1, \dots, X_n), n \geq 1\}$ satisfying

$$(1.1) P_\theta \{ \theta \in \bigcap_{n=1}^{\infty} I_n \} \geq 1 - \alpha$$

for a fixed α ($0 < \alpha < 1$) is a $(1 - \alpha)$ confidence sequence for θ . If m is a finite non-random integer and $I_n = (-\infty, \infty)$ for all $n > m$, then (1.1) becomes

$$(1.2) P_\theta \{ \theta \in \bigcap_{n=1}^m I_n \} \geq 1 - \alpha$$

and we speak of a finite confidence sequence. In the contrary case we have an infinite confidence sequence.

One of the main applications of finite confidence sequences is to interval estimation based on samples of random size, i.e. we wish to construct a $(1 - \alpha)$ confidence interval for θ on the basis of observations $X_1, \dots, X_N$ where N is an integer r.v. satisfying $N \leq m$ a.s.

Examples

(a) Let H_1 and H_2 be hypotheses concerning θ , and let $N \leq m$ be the stopping instant of a truncated sequential test of H_1 vs. H_2 . Apart from deciding between H_1 and

1. Work done while the author was visiting Potchefstroom University.

H_2 , the sequential test gives very little additional information concerning θ . The data at our disposal are the observations $X_1, \dots, X_N$ and from these we wish to construct a $(1 - \alpha)$ confidence interval for θ .

- (b) An experimenter draws a sample of m elements from a population. Due to circumstances beyond his control only N of the m elements provide suitable data. His confidence statements regarding an unknown parameter θ of the population will therefore have to be made on the basis of a sample of random size $N \leq m$.

If $\{I_n, n \leq m\}$ is a finite confidence sequence for θ , then it is easily seen from (1.2) that I_N is a $(1 - \alpha)$ confidence interval for θ . Thus finite confidence sequences provide us with one possible approach to the problem of interval estimation based on samples of random size.

In recent years a fair amount of attention has been devoted to the construction of infinite confidence sequences, see e.g. Lai (1976 (a) and (b)) and the references given there. The same cannot, however, be said about the finite case, the only work we are aware of being that of Paulson (1969).

Section 2 of this paper states some limit theorems which provide a method for constructing finite non-parametric confidence sequences which give approximate coverage probability $(1 - \alpha)$ for theoretically large values of m . In Section 3 the parameter θ represents the median of a distribution and confidence sequences based on the sign and Wilcoxon statistics are considered. Monte Carlo experiments indicate that the confidence sequences derived from the limit theorems of Section 2 actually satisfy (1.2) for $m = 20, (10), 50$ and $\alpha = 0,01; 0,05; 0,10$. In Section 4 the parameter θ represents the slope of a linear regression function and confidence sequences based on Kendall's tau statistic are considered. Here the Monte Carlo experiments indicate that the sequences derived from limit theorems violate (1.2) for $m = 20, \dots, 50$ and $\alpha = 0,01; 0,05; 0,10$. In both Sections 3 and 4 we empirically construct confidence sequences for which there is approximate equality in (1.2). Section 5 discusses some further questions relating to finite confidence sequences while Section 6 considers two examples of the application of infinite confidence sequences to problems of statistical inference.

2. Preliminary Results

Let $\{T_n, n \geq 1\}$ be a sequence of statistics, $Y_1, Y_2, \dots$, a sequence of i.i.d. r.v.'s and $g(t)$ a continuous function on $[0,1]$. Introduce the conditions C(1), C(2) and C(3) as follows:

$$C(1) \quad T_n = \sum_{i=1}^n Y_i + o(n^{1/2}) \text{ a.s. for } n \rightarrow \infty.$$

$$C(2) \quad EY_i = 0, EY_i^2 = 1,$$

$$C(3) \quad t^{-1/2} g(t) \uparrow \infty \text{ for } t \downarrow 0, \int_0^1 t^{-3/2} g(t) \exp(-g^2(t)/8t) dt < \infty$$

and let $\{W(t), t \geq 0\}$ denote a standard Wiener process on $[0, \infty)$.

Theorem 1

Let $\{r \equiv r(m)\}$ be a sequence of positive integers satisfying $1 \leq r \leq m$ for all m and $\lim_{m \rightarrow \infty} r/m = x$ with $0 \leq x < 1$. Then

(a) if $x = 0$:

$$(2.1) \quad \lim_{m \rightarrow \infty} P \{|T_n| < m^{1/2} g(n/m) \text{ for } n = r, r+1, \dots, m\}$$

$$= P \{|W(t)| < g(t) \text{ for all } t \in (0, 1]\}$$

provided $C(1)$, $C(2)$ and $C(3)$ hold.

(b) if $x > 0$:

$$(2.2) \quad \lim_{m \rightarrow \infty} P \{|T_n| < m^{1/2} g(n/m) \text{ for } n = r, r+1, \dots, m\}$$

$$= P \{|W(t)| < g(t) \text{ for all } t \in [x, 1]\}.$$

provided $C(1)$ and $C(2)$ hold.

Proof

(a) has been proved by Lombard (1976: Section 2) and (b) follows easily from Donsker's theorem.

For future reference we now define the following functions.

$$g_1(t, y) = yt^{1/2} \text{ for } y, t \geq 0$$

$$g_2(t, y) = y; g_3(t, y) = \{t(y - \log t)\}^{1/2} \text{ for } 0 < t \leq 1 \text{ and } y \geq 0.$$

$$g_4(t, y) = \{(t+1)(y + \log(t+1))\}^{1/2} \text{ for } t \geq 0 \text{ and } y \geq 0.$$

Let Φ and ϕ denote the distribution function and density function respectively of the standard normal distribution, and define the constants $|a|_2^\alpha$ and $|a|_3^\alpha$ by

$$\sum_{k=-\infty}^{\infty} (-1)^k \{ \Phi \{ (2k+1) |a|_2^\alpha \} - \Phi \{ (2k-1) |a|_2^\alpha \} \} = 1 - \alpha$$

and

$$1 - \Phi \{ (|a|_3^\alpha)^{1/2} \} + (|a|_3^\alpha)^{1/2} \phi \{ (|a|_3^\alpha)^{1/2} \} = \alpha/2.$$

Now g_2 and g_3 both satisfy C(3). Hence if C(1) and C(2) hold then it follows from (2.1) and Billingsley (1968), p. 79, that

$$\begin{aligned} (2.3) \quad 1 - \alpha &= \lim_{m \rightarrow \infty} P \{ |T_n| < m^{1/2} g_2(n/m, |a|_2^\alpha) \text{ for } n = r, r+1, \dots, m \} \\ &= \lim_{m \rightarrow \infty} P \{ |T_n| < m^{1/2} |a|_2^\alpha \text{ for } n = r, r+1, \dots, m \} \end{aligned}$$

and from (2.1) and Robbins & Siegmund (1970), p. 1414, that

$$\begin{aligned} (2.4) \quad 1 - \alpha &= \lim_{m \rightarrow \infty} P \{ |T_n| < m^{1/2} g_3(n/m, |a|_3^\alpha) \text{ for } n = r, r+1, \dots, m \} \\ &= \lim_{m \rightarrow \infty} P \{ |T_n| < \{ n(|a|_3^\alpha - \log n + \log m) \}^{1/2} \text{ for } n = r, r+1, \dots, m \} \end{aligned}$$

provided that $r = o(m)$ in both cases.

TABLE I

Values of $|a|_2^\alpha$ and $|a|_3^\alpha$

α	$* a _2^\alpha$	$ a _3^\alpha$
0,10	1,96	6,25
0,05	2,24	7,81
0,01	2,81	11,34

* These values were obtained by using the approximation

$$\sum_{k=-\infty}^{\infty} (-1)^k \{ \Phi \{ (2k+1) |a|_2^\alpha \} - \Phi \{ (2k-1) |a|_2^\alpha \} \} \approx 4\Phi(|a|_2^\alpha) - 3$$

which is obtained by neglecting in the series all terms except the three with indices $k = -1, 0$ and 1 .

Theorem 2

Let $a(x, \alpha)$ denote the solution in y of the equation

$$\alpha = (2/\pi)^{1/2} y \exp(-y^2/2) \{ \log x^{-1} - y^{-2} \log x^{-1} + 2y^{-2} \} / 2$$

Then

$$(2.5) \lim_{x \rightarrow 0} \lim_{m \rightarrow \infty} P \{ |T_n| < m^{1/2} g_1(n/m, a(x, \alpha)) \text{ for } n=r, r+1, \dots, m \} = 1 - \alpha$$

provided $\lim_{m \rightarrow \infty} r/m = x$ and $0 < x < 1$

Proof

Recall that if $\{W(t), 0 \leq t \leq 1\}$ is a standard Wiener process, and $U(t) = e^{-t} W(e^{2t})$, then $\{U(t), 0 \leq t \leq (\log x^{-1})/2\}$ is the stationary Ornstein-Uhlenbeck process with $EU(t) = 0$ and $EU(s)U(t) = \exp(-|t-s|)$. Together with part (b) of theorem 1 this leads to

$$\begin{aligned} & \lim_{x \rightarrow 0} \lim_{m \rightarrow \infty} P \{ |T_n| < m^{1/2} g_1(n/m, a(x, \alpha)) \text{ for } n=r, r+1, \dots, m \} \\ &= \lim_{x \rightarrow 0} P \{ |U(t)| < a(x, \alpha) \text{ for all } t \in [0, (\log x^{-1})/2] \} \\ &= 1 - \alpha, \end{aligned}$$

from the definition of $a(x, \alpha)$ and from (8) of Dirkse (1975), since $a(x, \alpha) \rightarrow \infty$ as $x \rightarrow 0$.

It seems reasonable to deduce from (2.5) that for m sufficiently large

$$(2.6) P \{ |T_n| < a(1/m, \alpha) n^{1/2} \text{ for } n = 1, 2, \dots, m \} \approx 1 - \alpha.$$

TABLE 2
Values of $a(x, \alpha)$

$x \quad \alpha$	0,01	0,05	0,10
1/50	3,543	2,994	2,712
1/40	3,525	2,973	2,688
1/30	3,501	2,943	2,655
1/20	3,463	2,897	2,603
1/5	3,278	2,672	2,348
1/4	3,234	2,619	2,289
1/3	3,168	2,542	2,205
1/2	3,039	2,395	2,052

3. Finite confidence sequences for the median of a distribution

Let $F(x)$ be a continuous c.d.f. with median zero and let $X_1, X_2, \dots$ be i.i.d. observations distributed according to the c.d.f. $F_\theta(x) = F(x-\theta)$ where θ is a finite (unknown) constant. Denote by P_θ the probability measure induced by F_θ on the sample space of the $X_1, X_2, \dots$. In this section we consider the construction of finite confidence sequences for θ when F is unknown. It is shown that such a sequence may be based on the sign statistic. If we assume that X is symmetric about zero, then such sequences may be based on a general class of one-sample rank order statistics, but for the sake of simplicity we shall consider only the Wilcoxon statistic.

The sign statistic is defined by

$$(3.1) \quad S_n = \sum_{i=1}^n \text{sign}(X_i)$$

and the Wilcoxon statistic by

$$(3.2) \quad W_n = (n+1)^{-1} \sum_{i=1}^n \sum_{j=1}^i \text{sign}(X_i + X_j)$$

$$\text{where sign}(u) = \begin{cases} 1 & \text{if } u > 0 \\ 0 & \text{if } u = 0 \\ -1 & \text{if } u < 0 \end{cases}$$

Now S_n trivially satisfies $C(1)$ if $\theta=0$, while it follows from Sen and Ghosh (1973) Theorem 2.4, that $\sqrt{3} W_n$ satisfies $C(1)$ and $C(2)$ if $\theta=0$, provided that $F(x) = 1-F(-x)$ for all x . In the following, whenever the Wilcoxon statistic is under discussion, it will be tacitly assumed that the r.v. under discussion is symmetrically distributed about zero.

For a nonnegative continuous function $g(t)$ on $[0,1]$ and for $2 \leq n \leq m, m > 2$ define:

$$(3.3) \quad \underline{a}(n,m) = \max \{1, [n/2 - m^{1/2} g(n/m)/2]\}; \bar{a}(n,m) = n+1 - \underline{a}(n,m)$$

$$(3.4) \quad \underline{b}(n,m) = \max \{1, [n(n+1)/4 - (m/12)^{1/2} g(n/m)]\}; \bar{b}(n,m) = n(n+1)/2 + 1 - \underline{b}(n,m)$$

where $[x] =$ largest integer $\leq x$. Let $X_{n,1} < \dots < X_{n,n}$ denote the order statistics of the observations $X_i, 1 \leq i \leq n$ and let $\tilde{X}_{n,1} < \dots < \tilde{X}_{n,n(n+1)/2}$ denote the order statistics of the $n(n+1)/2$ averages $(X_i + X_j)/2, 1 \leq i \leq j \leq n$.

Define

$$I_{n,m}^S = (X_{n,\underline{a}(n,m)}, X_{n,\bar{a}(n,m)})$$

and

$$I_{n,m}^W = (\tilde{X}_{n,\underline{b}(n,m)}, \tilde{X}_{n,\bar{b}(n,m)})$$

Then $\{I_{n,m}^S; n = 2, 3, \dots, m\}$ and $\{I_{n,m}^W; n = 2, 3, \dots, m\}$

are finite confidence sequences for θ with confidence coefficients

$$P_0 \{ |S_n| < m^{1/2} g(n/m) \text{ for } n = 2, 3, \dots, m \}$$

and

$$P_0 \{ \sqrt{3} |W_n| < m^{1/2} g(n/m) \text{ for } n = 2, 3, \dots, m \}$$

respectively. With any one of the choices $g(t) = g_1(t, a(1/m, \alpha))$, $g(t) = g_2(t, |a|_2^\alpha)$ or $g(t) = g_3(t, |a|_3^\alpha)$, it follows from (2.6), (2.3) or (2.4) that the confidence sequences should have coverage probability approximately equal to $(1-\alpha)$ for m sufficiently large. Define the statistics

$$S_{1,m} = \max_{2 \leq n \leq m} \{ |S_n| / n^{1/2} \}$$

$$S_{2,m} = \max_{2 \leq n \leq m} \{ |S_n| / m^{1/2} \}$$

$$S_{3,m} = \max_{2 \leq n \leq m} \{ S_n^2 / n + \log n - \log m \}$$

and let

$$\bar{F}(S_{i,m}; y) = P_0 \{ S_{i,m} \geq y \} \text{ for } i = 1, 2, 3.$$

Then the above theory indicates that, e.g. $1 - \bar{F}(S_{1,m}; a(x, \alpha)) \approx 1 - \alpha$ for m sufficiently large. Define the statistics $W_{i,m}$ by

$$W_{1,m} = \max_{2 \leq n \leq m} \{ \sqrt{3} |W_n| / n^{1/2} \}$$

$$W_{2,m} = \max_{2 \leq n \leq m} \{ \sqrt{3} |W_n| / m^{1/2} \}$$

$$W_{3,m} = \max_{2 \leq n \leq m} \{ 3W_n^2 / n + \log n - \log m \}$$

and let

$$\bar{F}(W_{i,m}; y) = P_0 \{ W_{i,m} \geq y \} \text{ for } i = 1, 2, 3.$$

The functions $\bar{F}(S_{i,m}; y)$ and $\bar{F}(W_{i,m}; y)$ for $i = 1, 2, 3$ were estimated by Monte Carlo methods as follows: Consider the estimation of $\bar{F}(S_{1,m}; y)$. Three thousand independent sets, each consisting of m independent random numbers from the uniform distribution on $[-1/2, 1/2]$, were generated. For each of these three thousand sets, the value of $S_{1,m}$ was calculated. The values of $S_{1,m}$ so obtained were grouped in a frequency table

with class intervals of length 0,25 and the estimate of $\bar{F}(S_{1,m};y)$ for $y = 0,0 (0,25)m^{1/2}$ obtained in the usual way. (An exception occurs in the case of $\bar{F}(S_{2,m};y)$ since the possible values of $m^{1/2}S_{2,m}$ are 0,1, ..., m. Here $\bar{F}(S_{2,m};y/m^{1/2})$ was estimated for $y = 0(1)m$). The values of m considered were $m = 20,30,40,50$.

TABLE 3

Monte Carlo estimates of $\bar{F}(S_{1,m};y)$.

$\begin{matrix} y \\ m \end{matrix}$	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50
20	0,321	0,263	0,097	0,053	0,020	0,008	0,003	0,001	0,000
30	0,354	0,253	0,114	0,064	0,027	0,014	0,007	0,004	0,001
40	0,385	0,288	0,132	0,072	0,028	0,014	0,004	0,001	0,000
50	0,388	0,282	0,144	0,086	0,040	0,020	0,006	0,003	0,001

TABLE 4

Monte Carlo estimates of $\bar{F}(S_{2,m};y/m^{1/2})$

$\begin{matrix} y \\ m \end{matrix}$	6	7	8	9	10	11	12	13
20	0,384	0,246	0,162	0,077	0,051	0,029	0,015	0,004
30	0,568	0,413	0,306	0,202	0,150	0,082	0,060	0,038
40	0,702	0,534	0,426	0,318	0,239	0,165	0,123	0,086
50	0,783	0,642	0,519	0,396	0,318	0,231	0,179	0,124

$\begin{matrix} y \\ m \end{matrix}$	14	15	16	17	18	19	20	21
20	0,002	0,000						
30	0,024	0,014	0,010	0,004	0,003	0,002	0,002	0,001
40	0,059	0,028	0,021	0,010	0,006	0,003	0,001	0,000
50	0,093	0,067	0,048	0,031	0,024	0,015	0,010	0,003

TABLE 5

Monte Carlo estimates of $\bar{F}(S_{3,m};y)$

$\begin{matrix} y \\ m \end{matrix}$	4,0	4,5	5,0	5,5	6,0	6,5	7,0	7,5	8,0	8,5	9,0	9,5
20	0,132	0,115	0,079	0,051	0,038	0,030	0,022	0,014	0,011	0,009	0,007	0,006
30	0,146	0,106	0,084	0,066	0,053	0,044	0,028	0,023	0,018	0,015	0,010	0,006
40	0,157	0,119	0,097	0,068	0,056	0,044	0,034	0,026	0,021	0,014	0,012	0,008
50	0,152	0,126	0,098	0,074	0,061	0,049	0,037	0,029	0,022	0,016	0,010	0,007

TABLE 6

Monte Carlo estimates of $\bar{F}(W_{1,m}y)$

$\begin{matrix} y \\ m \end{matrix}$	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50
20	0,259	0,143	0,077	0,032	0,010	0,004	0,001	0,000
30	0,311	0,187	0,092	0,042	0,019	0,009	0,003	0,001
40	0,357	0,207	0,110	0,056	0,025	0,009	0,003	0,000
50	0,374	0,223	0,122	0,056	0,025	0,012	0,005	0,002

TABLE 7

Monte Carlo estimates of $\bar{F}(W_{2,m}y)$

$\begin{matrix} y \\ m \end{matrix}$	1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00
20	0,357	0,213	0,120	0,069	0,034	0,012	0,006	0,002
30	0,365	0,229	0,131	0,066	0,034	0,016	0,008	0,004
40	0,382	0,228	0,135	0,072	0,041	0,023	0,009	0,004
50	0,384	0,229	0,135	0,071	0,035	0,018	0,007	0,003

TABLE 8

Monte Carlo estimates of $\bar{F}(W_{3,m}y)$

$\begin{matrix} y \\ m \end{matrix}$	4,5	5,0	5,5	6,0	6,5	7,0	7,5	8,0	8,5	9,0	9,5	10,0	10,5
20	0,144	0,109	0,079	0,058	0,046	0,032	0,027	0,017	0,011	0,005	0,002	0,001	
30	0,141	0,107	0,067	0,056	0,041	0,036	0,025	0,021	0,016	0,011	0,010	0,009	0,008
40	0,136	0,110	0,083	0,066	0,051	0,040	0,032	0,025	0,021	0,018	0,010	0,007	0,006
50	0,147	0,117	0,094	0,064	0,053	0,037	0,028	0,021	0,019	0,015	0,012	0,012	0,010

Now define for $i = 1, 2, 3$

$$|s|_{i,m}^{\alpha} = \inf \{y | \bar{F}(S_{i,m}y) \leq \alpha\}$$

and

$$|w|_{i,m}^{\alpha} = \inf \{y | \bar{F}(W_{i,m}y) \leq \alpha\}$$

Comparing Tables 1 and 2 with Tables 3 to 8, the indications are that

$$\bar{F}(S_{1,m}; a(1/m, \alpha)) < \bar{F}(S_{1,m}; |s|_{1,m}^{\alpha}),$$

$$\bar{F}(W_{1,m}; a(1/m, \alpha)) < \bar{F}(W_{1,m}; |w|_{1,m}^\alpha)$$

$$\bar{F}(S_{i,m}; |a|_i^\alpha) \leq \bar{F}(S_{i,m}; |s|_{i,m}^\alpha) \text{ for } i = 2, 3$$

$$\bar{F}(W_{i,m}; |a|_i^\alpha) < \bar{F}(W_{i,m}; |w|_{i,m}^\alpha) \text{ for } i = 2, 3.$$

The confidence sequences based on the functions $g_1(t, a(1/m, \alpha))$, $g_2(t, |a|_2^\alpha)$ and $g_3(t, |a|_3^\alpha)$ therefore all have coverage probabilities $\geq 1 - \alpha$ and may be quite safely used. Their width, however, will generally be somewhat greater than the widths of the corresponding sequences based on the function $g_i(t, |s|_{i,m}^\alpha)$ and $g_i(t, |w|_{i,m}^\alpha)$, $i = 1, 2, 3$. These last confidence sequences would therefore be preferred from a practical point of view. Tables of estimates of $|s|_{i,m}^\alpha$ and $|w|_{i,m}^\alpha$ (obtained by interpolation from Tables 3-8) are given below.

TABLE 9
Estimated values of $|s|_{i,m}^\alpha$.

$m \backslash \alpha$	$ s _{1,m}^\alpha$			$ s _{2,m}^\alpha$			$ s _{3,m}^\alpha$		
	0,01	0,05	0,10	0,01	0,05	0,10	0,01	0,05	0,10
20	2,71	2,27	1,91	2,91	2,24	2,01	8,13	5,55	4,72
30	2,83	2,35	2,07	2,92	2,37	2,01	9,00	6,15	4,67
40	2,83	2,38	2,13	2,69	2,37	2,06	9,17	6,29	4,95
50	2,93	2,44	2,19	2,83	2,26	1,98	9,00	6,45	4,97

TABLE 10
Estimated values of $|w|_{i,m}^\alpha$.

$m \backslash \alpha$	$ w _{1,m}^\alpha$			$ w _{2,m}^\alpha$			$ w _{3,m}^\alpha$		
	0,01	0,05	0,10	0,01	0,05	0,10	0,01	0,05	0,10
20	2,76	2,40	2,16	2,59	2,14	1,85	8,58	6,29	5,13
30	2,97	2,38	2,23	2,70	2,13	1,87	9,50	6,47	5,08
40	2,98	2,49	2,26	2,73	2,18	1,89	9,50	6,55	5,19
50	3,06	2,55	2,33	2,69	2,16	1,89	10,50	6,59	5,37

The question arises which of the functions $g_i(\cdot; \cdot)$ should be used when constructing the confidence sequences. For the values of α and m in Table 9 we see that

$$g_3(t, |s|_{3,m}^\alpha) > g_1(t, |s|_{1,m}^\alpha) \text{ for } t \in [0,1].$$

Furthermore, for the values of α in Table 9 and for $m = 20, 30, 40$ we have

$$g_1(t, |s|_{1,m}^\alpha) < g_2(t, |s|_{2,m}^\alpha) \text{ at least for } t \in [0,0; 0,90],$$

and it is clear that we would base our confidence sequences on $g_1(\cdot, \cdot)$. For $m = 50$ and $\alpha = 0,05; 0,10$; we have $g_1(t, |s|_{1,m}^\alpha) < g_2(t, |s|_{2,m}^\alpha)$ at least for $t \in [0,0; 0,8]$. In this case one could consider a compromise by basing the confidence sequence on the function $\min \{g_1(t, |s|_{1,m}^{\alpha/2}), g_2(t, |s|_{2,m}^{\alpha/2})\}$. Again, for the values of α and m in Table 10 we

see that $g_3(t, |w|_{3,m}^\alpha) > g_1(t, |w|_{1,m}^\alpha)$ for $t \in [0,1]$, with an exception at $m = 50, \alpha = 0,1$ where the last inequality holds for $t \in [0,0; 0,94]$.

Define

$$\begin{aligned} w_{m,\alpha} &= \sup \{t \in (0,1) \mid g_1(t, |w|_{1,m}^\alpha) \leq g_2(t, |w|_{2,m}^\alpha)\} \\ &= (|w|_{2,m}^\alpha / |w|_{1,m}^\alpha)^2 \end{aligned}$$

Some values of $w_{m,\alpha}$, estimated from Table 4, are given below.

TABLE 11
Estimated values of $w_{m,\alpha}$

α m	0,01	0,05	0,10
20	0,88	0,80	0,73
30	0,82	0,81	0,70
40	0,84	0,77	0,70
50	0,77	0,72	0,66

It is clear from Table 11 that neither the confidence sequence based on $g_1(\cdot, \cdot)$ nor the confidence sequence based on $g_2(\cdot, \cdot)$ gives minimum width over the whole range of values $n = 2, \dots, m$. Here again one could consider using a confidence sequence based on the function $\min \{g_1(t, |w|_{1,m}^{\alpha/2}), g_2(t, |w|_{2,m}^{\alpha/2})\} = \min \{|w|_{1,m}^{\alpha/2} t^{1/2}, |w|_{2,m}^{\alpha/2}\}$. The following table gives $|w|_{1,m}^\alpha t^{1/2}$, $|w|_{2,m}^\alpha$ and $g(m, t, \alpha/2) = \min \{|w|_{1,m}^{\alpha/2} t^{1/2}, |w|_{2,m}^{\alpha/2}\}$ for $m = 50$, $\alpha = 0,10$ and $t = 0,1$ (0,1) 1,0.

t	$ w _{1,m}^\alpha t^{1/2}$	$ w _{2,m}^\alpha$	$g(m,t,\alpha/2)$
0,1	0,74	1,89	0,81
0,2	1,04	1,89	1,14
0,3	1,28	1,89	1,40
0,4	1,47	1,89	1,61
0,5	1,65	1,89	1,80
0,6	1,80	1,89	1,98
0,7	1,95	1,89	2,13
0,8	2,08	1,89	2,16
0,9	2,21	1,89	2,16
1,0	2,33	1,89	2,16

Further calculation shows that $g(m,t,\alpha/2) < \max \{ |w|_{1,m}^\alpha t^{1/2}, |w|_{2,m}^\alpha \}$ for $t < 0,55$ and for $t > 0,85$, while the reverse inequality holds for $0,55 \leq t \leq 0,85$. The existence of this set of t -values is the main disadvantage associated with the "compromise" confidence sequence based on $g(m,t,\alpha/2)$.

4. Finite confidence sequences for a regression parameter.

Let F be a continuous c.d.f. and let $\{X_i, i \geq 1\}$ be independent r.v.'s. with c.d.f.'s $\{F_{\theta i}(x) = F(x - \alpha_i - \theta t_i), i \geq 1\}$ where $t_1 < t_2 < \dots$, are known constants and α and θ are unknown. Denote by P_θ the probability measure induced by the c.d.f.'s. $F_{\theta i}$ on the sample space of the sequence $\{X_i, i \geq 1\}$.

In this section we consider finite confidence sequences for θ based on the statistics

$$K_n = (n-1)^{-1} \sum_{i=1}^n \sum_{j=1}^i \text{sign}(X_i - X_j), \quad n \geq 2$$

For a nonnegative continuous function $g(t)$ on $[0,1]$ define, for $3 \leq n \leq m, m > 3$:

$\underline{c}(n,m) = \max \{1, [n(n-1)/4 - m^{1/2}g(n/m)/6]\}$; $\bar{c}(n,m) = n(n-1)/2 - \underline{c}(n,m) + 1$ and let $X_{n,1}^* < \dots < X_{n,n(n-1)/2}^*$ denote the order statistics of the divided differences $(X_i - \bar{X}_j)/(t_i - t_j), 1 \leq i < j \leq n$. If

$$I_{n,m}^K = (X_{n,\underline{c}(n,m)}^*, X_{n,\bar{c}(n,m)}^*)$$

then $\{I_{n,m}^K; 3 \leq n \leq m\}$ is a finite confidence sequence for θ with confidence coefficient

$$P_0 \{3|K_n| < m^{1/2}g(n/m) \text{ for } n = 3, 4, \dots, m\}$$

As far as we are aware, the statistic K_n does not satisfy condition C(1). Nevertheless, it follows from Theorem 2.3 of Lombard (1977) that (2.2) holds with $3|K_n|$ replacing $|T_n|$ and that (2.1) holds with $3|K_n|$ replacing $|T_n|$ in the case where $g(t)$ is a positive constant. It also follows that (2.3) and (2.6) remain valid with $3|K_n|$ replacing $|T_n|$. Consequently, with any of the choices $g(t) = g_1(t, a(1/m, \alpha))$ or $g(t) = g_2(t, |a|_2^\alpha)$, the resulting confidence coefficient is approximately $(1-\alpha)$ for m sufficiently large.

Define the statistics

$$K_{1,m} = \max_{3 \leq n \leq m} \{3|K_n|/n^{1/2}\}$$

and

$$K_{2,m} = \max_{3 \leq n \leq m} \{3|K_n|/m^{1/2}\}$$

and let

$$\bar{F}(K_{i,m}y) = P_0 \{K_{i,m} \geq y\}$$

The above theory indicates that $\bar{F}(K_{1,m}; a(x, \alpha)) \approx \alpha$ and $\bar{F}(K_{2,m}; |a|_2^\alpha) \approx \alpha$ for m sufficiently large. Using the same method as in Section 3 we have estimated the functions $\bar{F}(K_{1,m}y)$ and $\bar{F}(K_{2,m}y)$. The results are displayed below.

TABLE 12

Monte Carlo estimates of $\bar{F}(K_{1,m}y)$

$m \backslash y$	2,7	2,8	2,9	3,0	3,1	3,2	3,3	3,4	3,5	3,6	3,7	3,8	3,9
20	0,133	0,112	0,097	0,072	0,037	0,026	0,020	0,015	0,006	0,004	0,003	0,001	0,001
30	0,138	0,115	0,099	0,080	0,045	0,036	0,030	0,023	0,013	0,008	0,006	0,003	0,002
40	0,163	0,142	0,118	0,090	0,046	0,031	0,025	0,019	0,010	0,008	0,005	0,003	0,002
50	0,188	0,161	0,134	0,103	0,057	0,040	0,031	0,024	0,014	0,010	0,008	0,006	0,004

TABLE 13

Monte Carlo estimates of $\bar{F}(K_{2,m}y)$

$m \backslash y$	2,0	2,1	2,2	2,3	2,4	2,5	2,6	2,7	2,8	2,9	3,0	3,1	3,2
20	0,163	0,127	0,092	0,081	0,061	0,047	0,035	0,025	0,018	0,013	0,009	0,007	0,002
30	0,144	0,111	0,088	0,072	0,060	0,044	0,034	0,026	0,022	0,018	0,015	0,011	0,008
40	0,149	0,113	0,087	0,065	0,051	0,042	0,031	0,022	0,016	0,011	0,007	0,005	0,004
50	0,156	0,123	0,095	0,073	0,057	0,043	0,032	0,023	0,017	0,011	0,008	0,005	0,004

Now define

$$|\kappa|_{i,m}^{\alpha} = \inf \{y | \bar{F}(K_{i,m}y) \leq \alpha\}$$

Comparing Tables 1 and 2 with Tables 12 and 13 we see that

$$\bar{F}(K_{1,m}; a(1/m, \alpha)) > \bar{F}(K_{1,m}; |\kappa|_{1,m}^{\alpha})$$

and

$$\bar{F}(K_{2,m}; |a|_2^{\alpha}) > \bar{F}(K_{2,m}; |\kappa|_{2,m}^{\alpha}).$$

The confidence sequences based on the functions $g_1(t, a(1/m, \alpha))$ and $g_2(t, |a|_2^{\alpha})$ therefore both have confidence coefficients $< (1 - \alpha)$. This is in sharp contrast to what we found in Section 3 regarding the statistics S_n and $\sqrt{3} W_n$.

Estimates of $|\kappa|_{i,m}^{\alpha}$, $i = 1, 2$, obtained from table 6 are given below.

TABLE 14

Estimated values of $|\kappa|_{i,m}^{\alpha}$, $i = 1, 2$

α m	$ \kappa _{1,m}^{\alpha}$			$ \kappa _{2,m}^{\alpha}$		
	0,01	0,05	0,10	0,01	0,05	0,10
20	3,46	3,06	2,88	2,98	2,48	2,18
30	3,56	3,08	2,90	3,13	2,46	2,15
40	3,50	3,09	2,96	2,92	2,41	2,15
50	3,60	3,17	3,01	2,92	2,45	2,18

Define $\kappa_{m,\alpha} = \sup \{t \in (0,1] | g_1(t, |\kappa|_{1,m}^{\alpha}) < g_2(t, |\kappa|_{2,m}^{\alpha})\}$

$$= (|\kappa|_{2,m}^{\alpha} / |\kappa|_{1,m}^{\alpha})^2$$

TABLE 15

Estimated values of $\kappa_{m,\alpha}$

α m	0,01	0,05	0,10
20	0,74	0,67	0,57
30	0,77	0,64	0,55
40	0,70	0,61	0,53
50	0,70	0,60	0,52

From Table 15 it is once again clear that neither of the two confidence sequences has minimum width over the whole range of values of $n=3,4, \dots, m$.

5. Some General Comments

- (A) The discussion thus far concerned finite confidence sequences based on the test statistics T_n . Finite confidence sequences may however also be based on robust estimators. Let $X_1, X_2, \dots$ be i.i.d. random variables distributed according to the c.d.f. $F_\theta(x) = F(x - \theta)$ where θ is a finite (unknown) constant. Let $\hat{\theta}_n = \hat{\theta}_n(X_1, \dots, X_n)$ be a translation invariant point estimator of θ and suppose that C(1) holds with T_n replaced by $n(\hat{\theta}_n - \theta)$ where now $EY_i = 0$ and $E(Y_i^2) = \sigma^2$, an unknown positive constant. If, in addition, C(3) also holds, Swanepoel (1977, Theorem 2.2) has shown that Theorem 1 above remains valid if T_n is replaced by $n(\hat{\theta}_n - \theta)\sigma_n^{-1}$ where $\sigma_n = \sigma_n(X_1, \dots, X_n)$ is a location invariant point estimate of σ satisfying certain regularity conditions. Letting

$$I_{n,m} = (\hat{\theta}_n - m^{1/2}g(n/m)\sigma_n/n, \hat{\theta}_n + m^{1/2}g(n/m)\sigma_n/n)$$

we see that $\{I_{n,m}; n = 1, 2, \dots, m\}$ constitutes a finite confidence sequence for θ . Swanepoel (1977) has considered some choices of $\hat{\theta}_n$, e.g. where $\hat{\theta}_n$ is an L-estimator or M-estimator.

- (B) Let $\{I_{n,m}; n = 1, 2, \dots, m\}$ be a finite $(1-\alpha)$ confidence sequence for a parameter θ and suppose the r.v.'s $X_1, X_2, \dots$ are obtained sequentially. The intervals $I_{n,m}; n = 1, 2, \dots, m$ can therefore be computed sequentially. Letting $J_{n,m} = \bigcap_{k=1}^n I_{k,m}$ for $n = 1, 2, \dots, m$ we see that $\{J_{n,m}; n = 1, 2, \dots, m\}$ is also a finite $(1-\alpha)$ confidence sequence for θ and that $J_{1,m} \supset J_{2,m} \supset \dots \supset J_{m,m}$. Since

the length of $J_{n,m}$ therefore does not exceed that of $I_{n,m}$, the sequence $\{J_{n,m}\}$ will generally provide better estimates of θ than will $\{I_{n,m}\}$.

The sequence $\{J_{n,m}\}$ seems to have one disadvantage in that $J_{n,m}$ may become empty for some $n \leq m$. To examine this more closely we follow Geertsema et al. (1977) in defining:

$$\Omega_0 = \{ \omega : \theta \in I_{n,m} \text{ for } n = 1, 2, \dots, m \}$$

and

$$E = \{ \omega : J_{n,m} \text{ is empty for some } n = 1, 2, \dots, m \}$$

where $\omega = (X_1, X_2, \dots)$ denotes a typical sample point. Now $P_\theta \{E\} \leq P_\theta \{\Omega_0^c\} \leq \alpha$ so that E will rarely occur. But if E does occur, say $J_{n_0,m}$ is the first empty $J_{n,m}$ interval, then obviously at least one of the intervals $I_{1,m}, \dots, I_{n_0-1,m}$ does not contain θ . We cannot however identify the delinquent intervals, nor can we propose a general procedure to be followed in such a case. Paulson (1969) recommends that the procedure be repeated with fresh data. Suppose now that if E occurs, the statistician follows some procedure leading to an interval estimate I of θ . Defining

$$J = \begin{cases} J_{N,m} & \text{if } E \text{ does not occur} \\ I & \text{if } E \text{ does occur} \end{cases}$$

we see that

$$P_\theta \{ \theta \notin J \} = P_\theta \{ \theta \notin J_{N,m}, \Omega_0 \} + P_\theta \{ \theta \notin I, \Omega_0^c \} \leq \alpha$$

so that J is a $(1-\alpha)$ confidence interval for θ . Thus, no matter what interval the statistician uses in the event that E occurs the interval J defined above constitutes a $(1-\alpha)$ confidence interval for θ .

6. Infinite Confidence Sequences

It is possible, by using e.g. the inequalities of Robbins (1970), to obtain confidence sequences of the form $\{I_{n,m}; n \geq k\}$ where k is a fixed integer and m a positive constant. In this section we give two examples where particular choices of m lead to interesting solutions of some problems of statistical inference.

Example 1

(Fixed width sequential confidence interval for the mean of a normal distribution.)
Suppose the r.v.'s $X_1, X_2, \dots$ are i.i.d. $N(\theta, \sigma^2)$ with σ^2 unknown and that an inter-

val I is required, satisfying

$$(A) \quad P_{\theta} \{ \theta \in I \} \geq 1 - \alpha$$

and

$$(B) \quad \text{length } \{I\} \leq L,$$

where $L > 0$ and $0 < \alpha < 1$ are given. Let

$$\bar{X}_n = n^{-1} \sum_{i=1}^n X_i \quad \text{and} \quad \sigma_n^2 = (n-1)^{-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2, \quad n \geq 2$$

and define

$$N = \text{first } n \geq 3 \text{ at which } \sigma_n^2 < nL^2/4a^2$$

where $2\Phi(a) - 1 = \alpha$. Simons (1968) has shown that there exists a constant K (hitherto unknown) depending only on α such that if $N_* = N + K$, then the interval $I = (\bar{X}_{N_*} - L/2, \bar{X}_{N_*} + L/2)$ satisfies the requirements (A) and (B) above.

Furthermore, as $L \rightarrow 0$,

$$EN_* \sim 4\sigma^2 a^2 L^{-2} \quad \text{and} \quad \text{var}(N_* L^2) \rightarrow 0.$$

Earlier, Stein (1945) developed a two-stage procedure for obtaining an interval I satisfying (A) and (B). If M denotes the sample size required by Stein's procedure it holds that $EM = O(L^{-2})$ while $\text{var}(ML^2) \rightarrow C$, a positive constant.

We shall now develop a practicable sequential procedure with stopping rule N^* which yields an interval I satisfying (A) and (B) and such that

$$EN^* \sim L^{-2} \times (\text{constant}) \quad \text{and} \quad \text{var}(N^* L^2) \rightarrow 0.$$

Let $V_n^2 = \sigma_{n+1}^2$ for $n \geq 1$ and take

$$h_1(t) = g_4(t, -2\log(\alpha/2)) \quad \text{and} \quad h_2(t) = g_4(t, r^2(\alpha))$$

where $(1/2)e^{-r^2(\alpha)/2}/\Phi(r(\alpha)) = \alpha/2$. Then we obtain from (17) and (18) of Robbins (1970) (c.f. also Darling & Robbins (1967), p. 66) that

$$P_{\theta} \{ |\bar{X}_n - \theta| \leq n^{-1} m^{1/2} h_1(n/m) \sigma \text{ for } n = k, k+1, \dots \} \geq 1 - \alpha/2$$

and

$$P_{\theta} \{ \sigma \leq V_n \{ 1 - \sqrt{2} n^{-1} m^{1/2} h_2(n/m) \}^{-1/2} \text{ for } n = k, k+1, \dots \} \geq 1 - \alpha/2,$$

where $k \geq 1$ is a fixed integer constant to be specified later. The above inequalities imply

$$(6.1) \quad P_{\theta} \{ |\bar{X}_n - \theta| \leq n^{-1} m^{1/2} h_1(n/m) \{ 1 - \sqrt{2} n^{-1} m^{1/2} h_2(n/m) \}^{-1/2} V_n \text{ for } n = k, k+1, \dots \} \geq 1 - \alpha.$$

Now set $m = L^{-2}$, $f(t) = t^{-1} h_1(t)$ and $g(t) = t^{-1} h_2(t)$, and define

$$N^* = \text{first } n \geq k \text{ at which } 2f(nL^2) \leq \{ 1 - \sqrt{2} Lg(nL^2) \}^{1/2} V_n^{-1}$$

where $k \equiv k(L)$ is so large that $1 - \sqrt{2} Lg(nL^2) > 0$ for $n \geq k(L)$.

The interval I with endpoints $\bar{X}_{N^*} \pm Lf(N^*L^2) \{ 1 - \sqrt{2} Lg(N^*L^2) \}^{-1/2} V_{N^*}$ then satisfies both (A) and (B) (c.f. (6.1)).

Theorem 6.1

$EN^* \sim f^{-1}(1/2\sigma)L^{-2}$ where f^{-1} is the function inverse to f , and $\text{var}(N^*L^2) \rightarrow 0$ as $L \rightarrow 0$.

Proof

Since $k(L) = O(L^{-1})$ for $L \rightarrow 0$, we have $N^* \rightarrow \infty$ a.s. from the definition of N^* . Together with the inequality

$$(6.2) \quad 2f(N^*L^2) \leq \{ 1 - \sqrt{2} Lg(N^*L^2) \}^{1/2} V_{N^*}^{-1} \leq V_{N^*-1}^{-1}$$

this shows that $\liminf_{L \rightarrow 0} N^*L^2 > 0$ a.s. Since therefore $N^* > k(L)$ for all sufficiently small L , we have for all such L the inequality.

$$(6.3) \quad 2f((N^* - 1)L^2) > \{ 1 - \sqrt{2} Lg((N^* - 1)L^2) \}^{1/2} V_{N^*-1}^{-1},$$

from which we deduce that $\limsup_{L \rightarrow 0} N^*L^2 < \infty$ a.s.

We have therefore shown that, with probability one,

$$(6.4) \quad 0 < \liminf_{L \rightarrow 0} N^*L^2 \leq \limsup_{L \rightarrow 0} N^*L^2 < \infty.$$

Using (6.4) in (6.2) and (6.3) gives

$$\limsup_{L \rightarrow 0} f(N^*L^2) \leq 1/2\sigma \text{ and } \liminf_{L \rightarrow 0} f((N^*-1)L^2) \geq 1/2\sigma,$$

which in turn gives

$$(6.5) \lim_{L \rightarrow 0} N^*L^2 = f^{-1}(1/2\sigma)$$

An application of Lemma 3.2 of Bickel and Yahav (1968) shows that the set $\{(N^*L^2)^q\}_{0 < L \leq 1}$ is uniformly integrable for all $q > 0$. This, together with (6.5) proves the theorem.

Remark:

Let $b \equiv b(m, \alpha)$ be defined by $2eF_m(b) - 2e + 1 = 1 - \alpha$, where $F_m(\cdot)$ is the distribution function of a Student t random variable with m degrees of freedom, and let $N = \text{first } n \geq m + 1 \text{ at which } \sigma_n^2 < nL^2/4b^2$.

If I denotes the interval $(\bar{X}_N - L/2, \bar{X}_N + L/2)$, it has been shown by Starr and Woodroffe (1968) that requirements (A) and (B) are satisfied and $\lim_{L \rightarrow 0} ENL^2 = 4\sigma^2b^2$. Together with Theorem 6.1 this gives

$$\lim_{L \rightarrow 0} EN/EN^* = 4\sigma^2b^2/f^{-1}(1/2\sigma).$$

Now if $\alpha \downarrow 0$ it is easy to show that $b^2 \sim (\text{constant}) \times \alpha^{-m/2}$ and

$$f^{-1}(1/2\sigma) \sim 8(\log \alpha^{-1})\sigma^2,$$

with the result:

$$\lim_{\alpha \rightarrow 0} \lim_{L \rightarrow 0} EN/EN^* = \infty.$$

Example 2

(One-sided confidence sequence and test of power one for the scale parameter of an exponential distribution.)

Let $X_1, X_2, \dots$, be i.i.d.r.v.'s with common density function

$$f_{\sigma, \mu}(x) = \sigma^{-1} \exp \{-(x-\mu)/\sigma\} I(x \geq \mu), \sigma > 0,$$

where both μ and σ are unknown and $I(\cdot)$ denotes the indicator function. If the order statistics of $X_1, \dots, X_n$ are denoted by $X_{n1} < X_{n2} < \dots < X_{nn}$ then the maximum likeli-

hood estimator of σ may be expressed as

$$(6.6) \hat{\sigma}_n = n^{-1} \sum_{i=1}^n X_i - X_{n1} = n^{-1} \sum_{i=2}^n (n-i+1)(X_{ni} - X_{n(i-1)}).$$

We shall now derive the inequality

$$(6.7) P_{\sigma, \mu} \{ \sigma < \hat{\sigma}_n \{ (n-1)/n - m^{1/2} g((n-1)/m)/n \}^{-1} \text{ for } n = 2, 3, \dots \} \geq 1 - \alpha$$

where $m > 0$ is an arbitrary constant and $g(\cdot)$ a continuous Robbins-Siegmund function (c.f. Lai (1976b), p. 300 f.f.) satisfying:

$$(6.8) P \{ W(t) < g(t) \text{ for all } t > 0 \} = 1 - \alpha$$

Lai (1976b), p. 306, has derived an inequality similar to (6.7) for the case where μ is known.

Let $Y_1, Y_2, \dots$, be i.i.d. with density function $f_{1,0}(x)$. If we put $Z_i = 1 - Y_i$, then $E[\exp(yZ_1)] \leq \exp(y^2/2)$ for $y \geq 0$. The inequality (6.7) is now a direct consequence of the inequalities (1) and (11) of Robbins (1970) together with (6.8) and the following lemma.

Lemma 6.1

Let $X_1, X_2, \dots$, be i.i.d. with density function $f_{1,\mu}(x)$ and let $Y_1, Y_2, \dots$, be i.i.d. with density function $f_{1,0}(x)$. Define

$$S_n = \sum_{i=1}^n (n-i+1) (X_{ni} - X_{n(i-1)}) \text{ and } S_n^* = \sum_{i=2}^n Y_i \text{ for } n \geq 2.$$

Then the sequences $\{S_n, n \geq 2\}$ and $\{S_n^*, n \geq 2\}$ are identically distributed.

Proof

Since $X_{ni} - X_{n(i-1)} = (X_{ni} - \mu) - (X_{n(i-1)} - \mu)$ we may assume without loss of generality that $\mu = 0$. Let

$$f(t_n, t_{n+1}) = E \exp(-t_n S_n - t_{n+1} S_{n+1})$$

and

$$g(t_n, t_{n+1}) = E \exp(-t_n S_n^* - t_{n+1} S_{n+1}^*)$$

where t_n, t_{n+1} are arbitrary positive constants. We shall prove that

$f(t_n, t_{n+1}) = g(t_n, t_{n+1})$ from which the lemma will follow by induction. Using (6.6) we get

$$S_{n+1} = S_n + nX_{n1} + X_{n+1} - (n+1) \min(X_{n1}; X_{n+1})$$

and this, together with the fact that S_n and the vector $(X_{n1}; X_{n+1})$ are independent (c.f. Feller (1966), p. 18) implies that

$$(6.9) \quad f(t_n, t_{n+1}) = (1+t_n+t_{n+1})^{-n+1} E \exp(-t_{n+1} \{nX_{n1} + X_{n+1} - (n+1) \min(X_{n1}; X_{n+1})\})$$

also

$$\begin{aligned} (6.10) \quad & E \exp(-t_{n+1} \{nX_{n1} + X_{n+1} - (n+1) \min(X_{n1}; X_{n+1})\}) \\ &= E \exp(-t_{n+1} \{nX_{n1} - nX_{n+1}\}) I(X_{n+1} < X_{n1}) \\ &+ E \exp(-t_{n+1} \{-X_{n1} + X_{n+1}\}) I(X_{n+1} \geq X_{n1}) \\ &= E \exp(nt_{n+1}X_{n+1}) \int_0^\infty \exp\{-y(t_{n+1} + 1)\} dy \\ &+ E \exp(-t_{n+1}X_{n+1}) \int_0^{nX_{n+1}} \exp\{-y(1-t_{n+1}/n)\} \cdot dy \\ &= (1+t_{n+1})^{-1} E \exp(-nX_{n+1}) + \{n/(n-t_{n+1})\} [E \exp(-t_{n+1}X_{n+1}) \\ &\quad - E \exp(-nX_{n+1})] \\ &= (1+t_{n+1})^{-1} (1+n)^{-1} + \{n/(n-t_{n+1})\} [(1+t_{n+1})^{-1} - (1+n)^{-1}] \\ &= (1+t_{n+1})^{-1} \end{aligned}$$

Thus, from (6.9) and (6.10) we get

$$f(t_n, t_{n+1}) = (1+t_n+t_{n+1})^{-n+1} (1+t_{n+1})^{-1} = g(t_n, t_{n+1}).$$

Consider now the testing problem $H_0: \sigma \geq \sigma_0$ versus $H_1: \sigma < \gamma \sigma_0$ with $\gamma = 1-m^{-1/2}$ and $\sigma_0 > 0$.

Define

$$(6.11) \quad N = \begin{cases} \text{first } n \geq 2 \text{ at which } \hat{\sigma}_n \leq \sigma_0 \{((n-1)/n - m^{1/2} g((n-1)/m)/n\} \\ \infty \text{ if no such } n \text{ exists,} \end{cases}$$

where

$$(6.12) \quad g(t) = t + (\log \alpha^{-1})/2$$

and accept H_1 as soon as it is observed that $N < \infty$. H_0 is never accepted.

Now from Robbins (1970), p. 1403 it follows that the $g(t)$ of (6.12) satisfies (6.8) and so (6.7) implies that $P_{\sigma, \mu} \{\text{accept } H_1\} \leq \alpha$ for all μ and for all $\sigma \geq \sigma_0$. Since also $P_{\sigma, \mu} \{N < \infty\} = 1$ for each $\sigma < \gamma \sigma_0$, the test has power 1 against H_1 .

Theorem 6.2

For each $\sigma < \gamma \sigma_0$ and for all μ :

$$(6.13) \quad 1 \leq (1 - \sigma/\sigma_0 \gamma) E_{\mu, \sigma}(N) - \{\gamma(1 - \gamma)\}^{-1} (\log \alpha^{-1})/2 \leq 2$$

Proof

Let

$$N = \begin{cases} \text{first } n \geq 2 \text{ at which } \sigma \sum_{i=1}^n Y_i \leq \sigma_0 \{((n-1)\gamma - (1-\gamma)^{-1} (\log \alpha^{-1})/2\} \equiv c_n \\ \infty \text{ if no such } n \text{ exists.} \end{cases}$$

Then N^* and N are identically distributed, c.f. Lemma 6.1. Now let $N_k^* = \min(N^*, k)$ for $k \geq 3$. On the event $\{N^* \geq 3\}$ we have from the definition of N^* :

$$\sigma \sum_{i=2}^{N_k^*} Y_i > \sigma \sum_{i=2}^{N_k^*-1} Y_i > c_{N_k^*-1}$$

while on the event $\{N^* = 2\}$ we have

$$\sigma \sum_{i=2}^{N_k^*} Y_i = \sigma Y_2 > c_{N_k^*-1}$$

since then $c_{N_k^*-1} = c_1 < 0$, while $Y_2 > 0$. Consequently, on the event $\{N^* \geq 2\}$

$$\sigma \sum_{i=2}^{N_k^*} Y_i > \sigma_0 (N_k^* - 2) \gamma - \sigma_0 (1 - \gamma)^{-1} (\log \alpha^{-1})/2.$$

Taking expectations on both sides of this inequality, applying Wald's Lemma to the left-hand side and then letting $k \rightarrow \infty$, we obtain the upper bound in (6.13). The lower bound follows similarly upon use of the inequality

$$\sigma \sum_{i=2}^{N^*} Y_i \leq c_{N^*}.$$

Define N as in (6.11) with

$$(6.14) \quad g(t) = g_A(t, a^2); t > 0 \text{ and } e^{-a^2/2}/\Phi(a) = 2\alpha.$$

This $g(t)$ is a Robbins-Siegmund function which satisfies (6.8), (c.f. Robbins (1970), p. 1403) and N now yields a test of power one for the testing problem $H_0: \sigma \geq \sigma_0$ versus $H_1: \sigma < \sigma_0$.

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